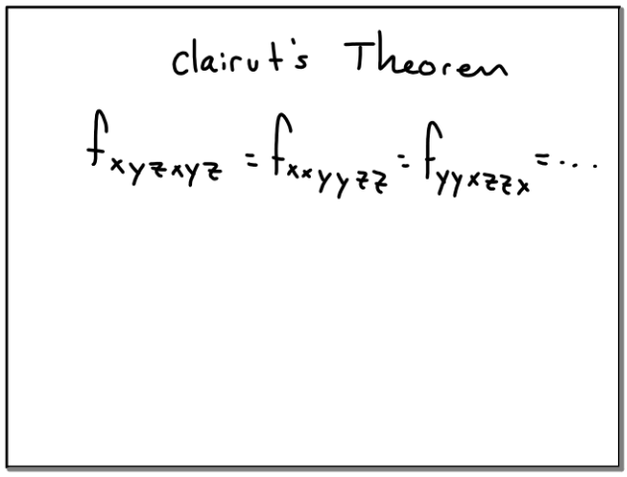
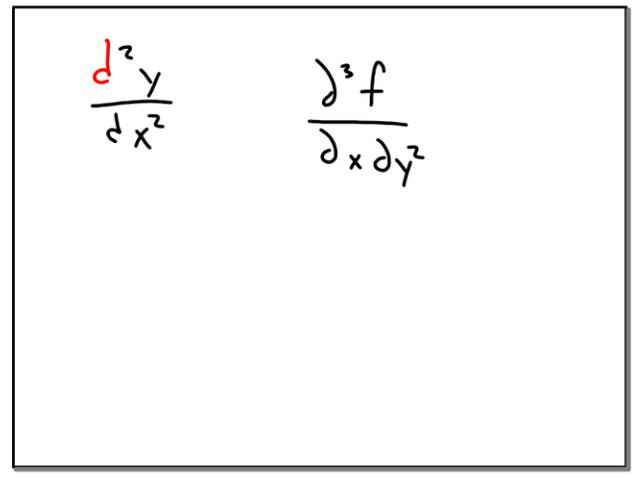
Ameet Rao's Math Notes

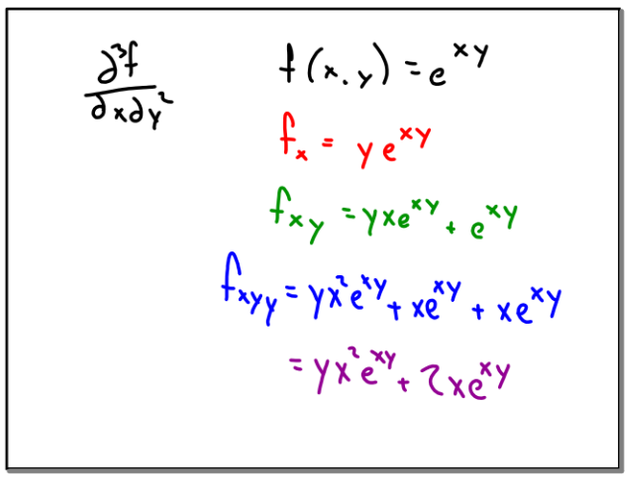
Higher-Order Partials, Clairaut's Theorem, the Chain Rule, and Implicit Differentiation



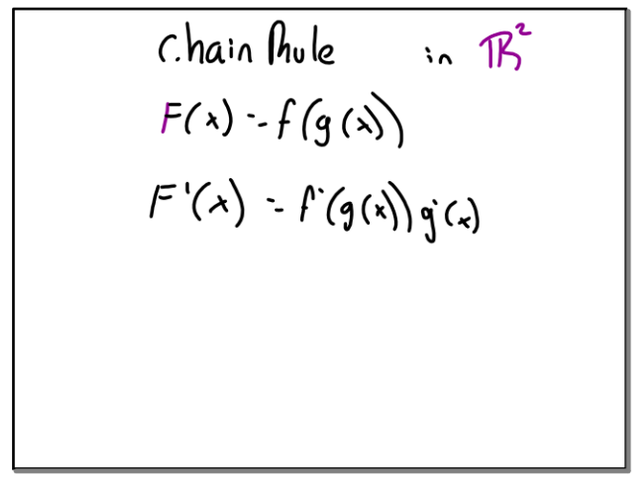
Clairaut's Theorem: if the mixed partials are continuous, the order you differentiate in doesn't matter. So f_xy = f_yx, and for higher orders f_xyzxyz = f_xxyyzz = f_yyxzzx: only how many times you hit each variable counts.



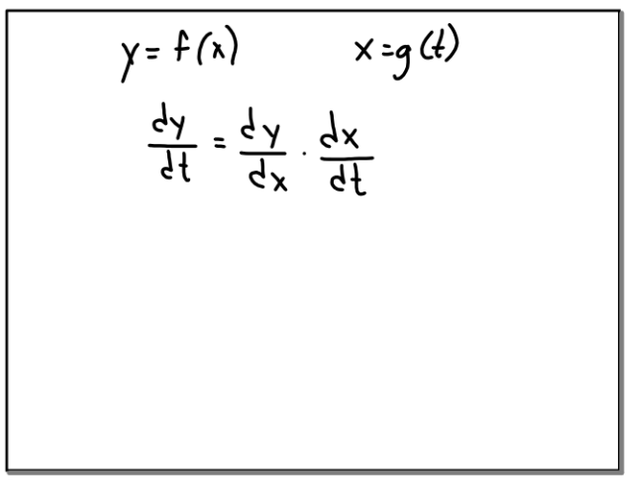
Notation: d²y/dx² is an ordinary second derivative. ∂³f/∂x∂y² is a third-order partial: differentiate with respect to x once and y twice. By Clairaut, the order doesn't change the answer.



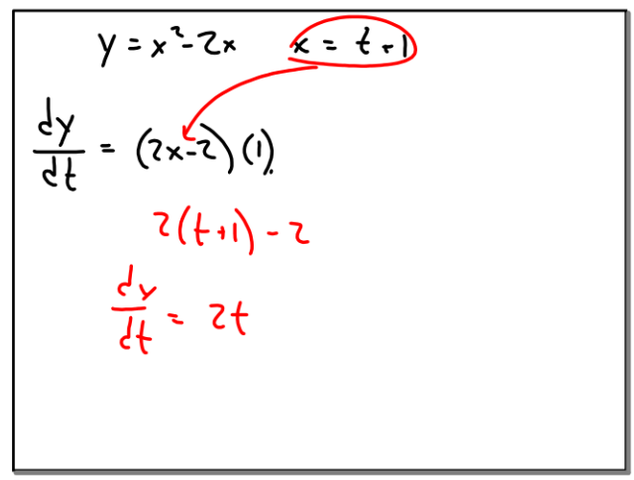
f = e^(xy), one variable at a time: f_x = ye^(xy); then f_xy = xye^(xy) + e^(xy) (product rule); then f_xyy = x²ye^(xy) + xe^(xy) + xe^(xy) = x²ye^(xy) + 2xe^(xy).



Review of the one-variable chain rule: if F(x) = f(g(x)), then F′(x) = f′(g(x))·g′(x). Derivative of the outside, evaluated at the inside, times the derivative of the inside.



The same rule in Leibniz form: if y depends on x and x depends on t, then dy/dt = (dy/dx)(dx/dt). The dx's look like they cancel, which is an easy way to remember it.



Example: y = x² − 2x with x = t + 1. dy/dt = (2x − 2)(1); substitute x = t + 1 to get 2(t + 1) − 2 = 2t. Always finish by writing the answer in terms of t.

Produced in 2026

$$z = f(x, y) \qquad x = g(t) \qquad y = h(t)$$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$

Chain rule for z = f(x, y) when x and y both depend on t: dz/dt = (∂z/∂x)(dx/dt) + (∂z/∂y)(dy/dt). One term for each path from z down to t, added together.

$$z = xe^{xy} \qquad x = t^2 \qquad y = \frac{1}{t} = t^{-1}$$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}$$

$$= \left(xye^{xy} + e^{xy}\right)(2t) + \left(x^2e^{xy}\right)\left(-t^{-2}\right)$$

$$= \left((t^2)(t^{-1})e^{(t^2)(t^{-1})} + e^{(t^2)(t^{-1})}\right)(2t) + \left((t^2)^2 e^{(t^2)(t^{-1})}\right)\left(-t^{-2}\right)$$

$$= \left(te^t + e^t\right)(2t) + \left(-t^2e^t\right)$$

$$\frac{dz}{dt} = t^2e^t + 2te^t$$

z = xe^(xy) with x = t², y = 1/t. Partials: z_x = xye^(xy) + e^(xy), z_y = x²e^(xy); dx/dt = 2t, dy/dt = −t⁻². Substitute (note xy = t) and simplify: dz/dt = (te^t + e^t)(2t) − t²e^t = t²e^t + 2te^t.

$$z = (t^2)^{e^{(t^2)(t^{-1})}}$$

$$z = t^2e^t$$

$$\frac{dz}{dt} = t^2e^t + 2te^t$$

Check by substituting first: z = t²·e^((t²)(1/t)) = t²e^t, and the product rule gives dz/dt = t²e^t + 2te^t. Same answer as the chain rule, which makes this a good way to check your work.

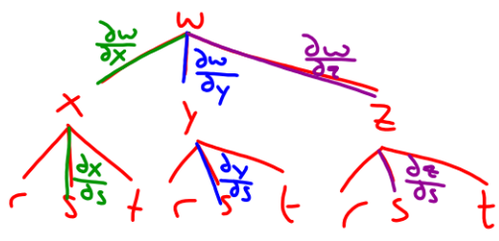
$$z = x^2y^3 + y\cos(x) \qquad x = \ln(t^2) \qquad y = \sin(4t)$$

Find $\frac{dz}{dt}$

Practice (not worked on the slide): with x = ln(t²) and y = sin 4t, dx/dt = 2/t and dy/dt = 4cos 4t. So dz/dt = (2xy³ − y sin x)(2/t) + (3x²y² + cos x)(4cos 4t).

$$w = f(x, y, z) \qquad x = g(r, s, t) \qquad y = h(r, s, t) \qquad z = k(r, s, t)$$

$$\frac{dw}{ds} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial s} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial s}$$



More variables: w = f(x, y, z) where x, y, z each depend on r, s, t. Use the tree diagram: follow every branch from w down to s, multiply along each branch, then add. ∂w/∂s = w_x·x_s + w_y·y_s + w_z·z_s.

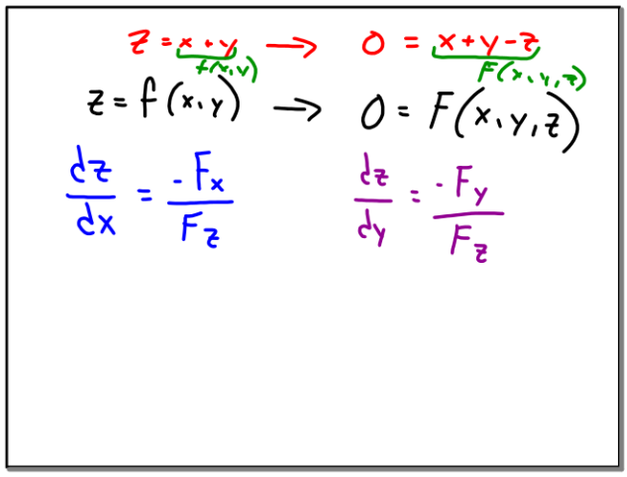
Implicit

$$y = g(x) \qquad z = f(x, y)$$

$$\frac{dz}{dx} = \frac{\partial z}{\partial y} \cdot \frac{dy}{dx} + \frac{\partial z}{\partial x} \cdot \frac{dx}{dx}$$

Implicit setup: if z = f(x, y) and y = g(x), then x affects z directly and through y. dz/dx = (∂z/∂x)(dx/dx) + (∂z/∂y)(dy/dx), and dx/dx = 1.

Produced in 2026



Implicit differentiation shortcut: move everything to one side, F(x, y, z) = 0 (e.g. z = x + y becomes x + y − z = 0). Then ∂z/∂x = −F_x/F_z and ∂z/∂y = −F_y/F_z. Check: F = x + y − z gives ∂z/∂x = −1/(−1) = 1, as expected.

Produced in 2026